

## **SUBJECT NAME: ROAD SAFETY ENGINEERING (BCV755A)**

### **MODULE:1**

**Syllabus: Accident Investigations and Risk Management:** Collection Of Accident Data, Assessment Of Road Safety, Methods To Identify And Prioritize Hazardous Locations And Elements, Determine Possible Causes Of Crashes, Crash Reduction Capabilities And Countermeasures, Effectiveness Of Safety Design Features, Accident Reconstruction, Condition And Collision Diagram.

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### **Introduction:**

The accident studies pertain to the road accidents that occur from time to time on an existing road system, though road accidents cannot be totally prevented, the rate can be decreased substantially by suitable traffic engineering and management measures.

**Objectives of Accident studies,**

The various objectives of the accident studies may be listed as:

- (a) to study the causes of accidents and to suggest corrective treatment at potential locations
- (b) to evaluate existing design, regulation and control measures
- (c) to support proposed changes in design, regulation and control measures in the selected zone
- (d) to carry out 'before and after studies' after implementing changes and to demonstrate the improvement in the accident problem
- (e) to make computations of financial loss due to accidents and
- (f) to provide economic justification for the improvement measures suggested by the traffic engineer

### **Collection Of Accident Data:**

Collection of accident is the first step in the accident studies, the details to be collected are briefly mentioned below,

- (a) General - date, time, persons involved in the accident and their particulars, classification of accident like fatal, serious, minor, property damage only, etc
- (b) Location - description and details of the location of accident supported by diagrams
- (c) Details of vehicles involved - registration number, make and description of the vehicles, loading details, vehicular defects
- (d) Nature of accident - condition of vehicles involved, details of collision, and pedestrians or objects involved, damages, injuries, casualty, etc

- (e) **Road condition** - details of road geometrics, whether the road stretch is straight or curved, surface characteristics such as dry, wet or slippery, etc
- (f) **Traffic condition** - type vehicles in the traffic flow, traffic volume and density, etc
- (g) **Primary causes of accident** - various possible causes and the primary cause of the accident
- (h) **Other probable causes/secondary and contributing causes** for the accident
- (i) **Accident cost** - the total cost of the accident, such as property damages, personal injuries and casualties, computed in terms of Rupees

### Methods for collection of accident data,

Collecting accident data is crucial for analyzing road safety, designing improvements, and forming policies. Below are the **main methods for collection of accident data**:

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#### 1. Police Records

- **Description:** Accidents reported to and recorded by local police departments.
- **Details Captured:** Time, location, vehicle type, driver info, road condition, cause, injuries, fatalities.
- **Advantages:** Official, standardized, and legally recognized.
- **Limitations:** Underreporting of minor accidents.

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#### 2. Hospital & Medical Records

- **Description:** Data from emergency rooms and trauma centers.
- **Details Captured:** Nature and severity of injuries, treatment, patient demographics.
- **Advantages:** Captures unreported accidents; useful for injury severity analysis.
- **Limitations:** May lack accident location and context.

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#### 3. Insurance Company Reports

- **Description:** Claims submitted for damages or medical treatment.
- **Details Captured:** Vehicle damage, cost, nature of impact.
- **Advantages:** Includes both minor and major accidents.
- **Limitations:** Data access is often private/confidential.

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#### 4. On-site Investigations

- **Description:** Detailed examination at the accident location by traffic engineers or researchers.

- **Details Captured:** Road geometry, skid marks, vehicle final positions, environmental conditions.
  - **Advantages:** Accurate and comprehensive.
  - **Limitations:** Time-sensitive; resource-intensive.
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## 5. Surveys & Interviews

- **Description:** Questionnaires or verbal interviews with drivers, victims, or witnesses.
  - **Details Captured:** Human factors, perception, pre-accident behavior.
  - **Advantages:** Provides subjective insights.
  - **Limitations:** Recall bias; limited sample size.
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## 6. CCTV / Video Surveillance

- **Description:** Traffic camera or surveillance footage from accident-prone areas.
  - **Details Captured:** Real-time visuals of accident sequence.
  - **Advantages:** Verifiable evidence; helps analyze crash dynamics.
  - **Limitations:** Limited coverage; privacy concerns.
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## 7. Automatic Traffic Recorders (ATRs) & IoT Sensors

- **Description:** Smart sensors embedded in road infrastructure or vehicles.
  - **Details Captured:** Speed, traffic volume, impact forces.
  - **Advantages:** Continuous monitoring; no human bias.
  - **Limitations:** High cost; data integration challenges.
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## 8. GPS & Telematics Data

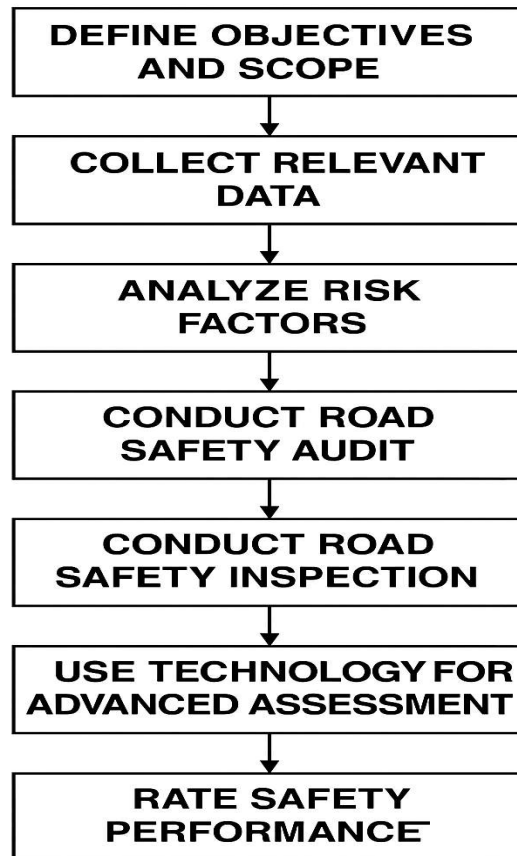
- **Description:** From fleet management systems, apps, or black boxes.
  - **Details Captured:** Vehicle speed, location, sudden stops, crash detection.
  - **Advantages:** Precise and real-time.
  - **Limitations:** Privacy and data ownership issues.
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## 9. Academic and Government Research Databases

- **Examples:**
  - India: **MoRTH accident database, IRAD (Integrated Road Accident Database)**
  - US: **NHTSA's FARS (Fatality Analysis Reporting System)**

## Assessment of Road Safety:

Road safety refers to the measures, practices, and strategies designed to prevent road users—such as drivers, passengers, pedestrians, and cyclists—from being killed or seriously injured in traffic accidents. It involves the safe design and operation of roads, vehicles, and traffic systems, as well as the enforcement of laws and promotion of responsible user behavior. The following flow chart depict how to conduct Road Safety Assessment.



Step-by-step procedure to assess

### **Step 1: Define Objectives and Scope**

**Objective:** Determine *why* and *where* the road safety assessment is being done.

- **Scope:** Is it a full corridor, a junction, a rural road, or an urban street?
- **Purpose:**
  - Reduce fatalities and serious injuries.
  - Evaluate post-construction performance.
  - Improve pedestrian or cyclist safety.
  - Conduct feasibility for road widening or signal changes.

**Example:** Assess the safety of a national highway stretch between two towns due to rising crash rates.

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### **◆ Step 2: Collect Relevant Data**

**Collect accurate and recent data** related to road use, crash occurrences, and road environment.

### 1. Crash/Accident Data

- Source: Police, hospitals, insurance, Integrated Road Accident Database (IRAD – India).
- Fields: Date, time, type of accident, location (GPS or chainage), weather, fatalities, injuries.

### 2. Traffic Data

- Volume (AADT – Annual Average Daily Traffic).
- Speed studies (average speed, 85th percentile speed).
- Vehicle mix (2W, 4W, buses, trucks).

### 3. Road Inventory

- Road width, number of lanes, shoulder condition, pavement condition.
- Signage, markings, lighting, intersections, curves, visibility.
- Nearby land use (schools, shops, bus stops).

### 4. Human Factors

- Driver behavior: speeding, drunk driving, distraction.
- Pedestrian movements.
- Helmet/seatbelt usage rates.

**Tools:** Traffic counters, GPS mapping, drone surveys, field notes.

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## ◆ Step 3: Analyze Risk Factors

**Goal:** Identify hazardous conditions or risky behaviors contributing to accidents.

- **Blackspot Identification:** Locations with high crash density or crash severity index.
- **Crash Pattern Analysis:**
  - Repeated types (e.g., rear-end, side-impact, pedestrian hits).
  - Time pattern (e.g., night crashes due to poor lighting).
- **Design Deficiencies:** Poor horizontal alignment, steep slopes, inadequate shoulder width, wrong signage.

**Tools:** GIS, crash plotting software, Excel heat maps, iRAP ViDA.

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## ◆ Step 4: Conduct Road Safety Audit (RSA)

A **systematic and formal safety review** done by an independent, qualified team at different stages:

- **Planning/Design Stage:** Ensure safety is built into drawings.
- **Construction Stage:** Check temporary traffic control and safety during work zones.
- **Pre-opening Stage:** Evaluate final road conditions before opening.
- **Post-opening Stage:** Assess real-world performance.

**Checklist Items:**

- Are signs visible at night?
- Is the road shoulder stable?
- Are pedestrian crossings adequate?

**Standard:** IRC SP 88-2019 (Road Safety Audit Manual – India)

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## ◆ Step 5: Conduct Road Safety Inspection (RSI)

An **on-site field visit** to visually assess the condition of an existing road from a safety perspective.

- **Visual checks** (no need for crash data).
- Focuses on:
  - Pavement defects

- Faded signs/markings
- Obstructions near the carriageway (trees, poles)
- Wrongly placed bus stops
- Poor drainage causing waterlogging

**Frequency:** Can be done periodically (monthly/quarterly).

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### ◆ Step 6: Use Technology for Advanced Assessment

Technology helps in **quantifying risks** and **predicting** unsafe conditions.

*Tools:*

- **CCTV & Drone Video Analytics:**
  - Study near misses, red-light violations, pedestrian behavior.
- **Crash Simulation Models:**
  - Software like VISSIM, SIDRA, PTV for dynamic modeling.
- **GIS Mapping:**
  - Visualizing crash hotspots and traffic patterns.
- **iRAP Star Rating System:**
  - Roads are rated from 1 to 5 stars based on risk exposure.

**Use Case:** Use drone footage to observe pedestrian conflict zones at a busy junction.

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### ◆ Step 7: Rate Safety Performance

Based on collected data, audits, and observations, assign a **safety score or rating**.

*Common Performance Indicators:*

- Crash rate per million vehicle-km
- Fatality and injury rate
- Helmet/seatbelt usage rate
- Response time to accidents
- iRAP star rating (1 star = very dangerous, 5 stars = very safe)

**Outcome:** Helps compare safety levels of different roads or the effect of implemented measures.

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### ◆ Step 8: Recommend Improvements

Based on identified risks and safety rating:

*A. Engineering Measures*

- Speed calming: rumble strips, speed humps, roundabouts.
- Signage upgrades, pedestrian footpaths, lighting.
- Guardrails, medians, shoulder strengthening.

*B. Enforcement*

- Install speed/red-light cameras.
- Conduct regular drunk driving checks.
- Deploy traffic police at blackspots.

*C. Education*

- Campaigns for helmet/seatbelt use.
- School zone awareness programs.

*D. Emergency Response*

- Improve ambulance access routes.
- GPS tracking for emergency services.

## Key performance indicators used in road safety assessment AND ITS influence road safety planning and policy development

Key Performance Indicators (KPIs) in road safety are essential tools used to assess the effectiveness of traffic management systems and safety interventions. These indicators provide measurable values that help authorities monitor progress, identify problem areas, and guide policy decisions. By evaluating various aspects of road safety performance, KPIs contribute to data-driven planning and continuous improvement in reducing road traffic injuries and fatalities.

**1. Fatality Rate:** One of the most widely used KPIs is the number of fatalities per 100,000 population or per billion vehicle kilometers traveled (VKT). It reflects the severity of traffic collisions and helps in international comparisons. A declining fatality rate is a strong sign of improved road safety.

**2. Serious Injury Rate:** This measure the number of people seriously injured in road traffic accidents, often standardized per 100,000 population or per billion VKT. Unlike fatalities, serious injuries may not always be reported consistently, but thi

**3. Crash Frequency and Rate:** This includes total road crashes over a specific period or within a certain area. It can be used to identify crash-prone locations, commonly referred to as “black spots.” Crash rate per VKT helps adjust for variations in traffic volume.

**4. Road User Behavior Indicators:** These include rates of seatbelt use, helmet use, compliance with speed limits, and levels of drunk or distracted driving. Data is collected through field surveys, roadside checks, and enforcement statistics. These indicators assess how well road users adhere to safety norms.

**5. Emergency Response Time:** This KPI measures the time taken for emergency services to arrive at accident scenes. Faster response times can significantly reduce fatalities and the severity of injuries, especially in critical trauma cases.

**6. Road Infrastructure Safety Rating:** Through tools like the International Road Assessment Programme (iRAP), roads are rated based on their safety features. This indicator highlights the need for infrastructure improvements in high-risk areas.

**7. Number of Road Safety Inspections and Audits:** This measures how often safety assessments are conducted on roads, especially during the design and post-construction phases. It reflects the proactive efforts of authorities in identifying and mitigating risks.

**Influence on Road Safety Planning and Policy Development:** KPIs directly influence road safety planning by highlighting trends and identifying priority areas for intervention. For example, if the fatality rate remains high despite reduced crash numbers, it may suggest that crashes are more severe, prompting the need for better vehicle safety standards or medical response systems.

Policymakers rely on KPIs to evaluate the success of existing strategies and to justify new policies. For instance, a low rate of seatbelt usage could lead to targeted enforcement campaigns or legislation mandating stricter penalties. Additionally, international road safety goals, such as those set by the UN Decade of Action for Road Safety, often require countries to track KPIs to ensure accountability and progress.

KPIs provide a structured approach to assessing and improving road safety. They enable evidence-based decision-making, enhance transparency, and guide the development of policies that save lives and prevent injuries on the roads.

## Determine Possible Causes Of Crashes:

There are five basic elements in a traffic accident, namely:

- (a) Road user
- (b) Vehicle
- (c) Road and its condition
- (d) Traffic
- (e) Environmental factors such as weather, visibility, etc.

Various causes of accidents may be listed as given below:

- (a) *Drivers:* Excessive speed and rash driving, carelessness, violation of rules and regulations, failure to see or understand the traffic situation, sign or signal, temporary effects due to fatigue, sleep or effect of consuming alcohol
- (b) *Pedestrians:* Violating regulations, carelessness while using the carriageway meant for vehicular traffic
- (c) *Passengers:* Alighting from or getting into moving vehicles
- (d) *Vehicle defects:* Failure of brakes, steering system, or lighting system, tyre burst and any other defect in the vehicles
- (e) *Road condition:* Slippery or skidding road surface, pot holes, ruts and other damaged conditions of the road surface, temporary obstruction to line of sight (caused by branch of tree or disabled vehicle) resulting in reduction in normal sight distance
- (f) *Road design:* Defective geometric design like inadequate sight distance at horizontal or vertical curves, improper curve design, inadequate width of shoulders, improper lighting and improper traffic control devices
- (g) *Traffic condition:* Other vehicles of the traffic stream, such as a vehicle moving ahead getting involved in accident, presence of disabled vehicle on the roadway
- (h) *Weather:* Unfavourable weather condition like mist, fog, snow, dust, smoke or heavy rainfall which restrict normal visibility and render driving unsafe
- (i) *Animals:* Stray animals on the road
- (j) *Other causes:* Incorrect signs or signals, gate of level crossing not closed when required, ribbon development, badly located advertisement boards or service station, etc.

## Methods To Identify And Prioritize Hazardous Locations And Elements:

### 1. Crash Frequency Method

- **What it does:** Identifies locations based on the number of crashes in a given period (typically 3–5 years).
  - **Identification Criteria:** Locations with crash counts above a predefined threshold are considered hazardous.
  - **Limitation:** Doesn't consider exposure (traffic volume) or crash severity.
  - **Prioritization:** Sites with the highest number of crashes are given priority.
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## 2. Crash Rate Method (Exposure-Based)

- **What it does:** Normalizes crash frequency by traffic volume and road length.
- **Formula:**

$$\text{Crash Rate} = \frac{\text{Crashes} \times 10^6}{\text{AADT} \times \text{Road Length} \times 365}$$

- **Benefit:** Allows fair comparison between roads with different traffic volumes.
  - **Prioritization:** Locations with unusually high crash rates are prioritized even if total crashes are low.
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## 3. Crash Severity Index Method

- **What it does:** Assigns weights to crashes based on severity (e.g., fatal = 5, serious = 3, minor = 1).
- **Formula:**

$$\text{Severity Index} = \frac{\text{Weighted Sum of Crashes}}{\text{Total Number of Crashes}}$$

- **Benefit:** Highlights locations with more serious crashes.
  - **Prioritization:** Sites with higher severity index scores are ranked higher.
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## 4. Crash Cost Analysis

- **What it does:** Calculates economic losses associated with crashes (injuries, fatalities, property damage).
  - **Application:** Helps in cost-benefit analysis for proposed safety improvements.
  - **Prioritization:** Locations causing higher societal costs are treated with urgency.
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## 5. Statistical Models (Poisson, Negative Binomial, Empirical Bayes)

- **What they do:**
  - Use mathematical models to predict expected crashes based on road characteristics.
  - Compare observed vs. expected crash values.
- **Empirical Bayes:** Combines historical data with predictions for better accuracy.

- **Prioritization:** Locations with the largest positive difference (Observed > Expected) are critical.

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## 6. Road Safety Audits (RSA)

- **What it does:** Independent safety review of existing/planned roads by a multidisciplinary team.
- **Focus:** Not just crash data, but also geometric design, visibility, signage, etc.
- **Prioritization:** Based on identified safety concerns and expert judgment.

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## 7. Traffic Conflict Technique (TCT)

- **What it does:** Observes near-miss situations as a proxy for potential crashes.
- **Use:** Proactive method where crash data is unavailable or outdated.
- **Prioritization:** Sites with the most frequent or severe conflicts are prioritized.

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## 8. GIS-Based Spatial Analysis

- **What it does:** Uses GIS tools to map crash locations and identify clusters/hotspots.
- **Methods:** Heat maps, kernel density estimation, spatial autocorrelation.
- **Prioritization:** High-density crash zones are prioritized based on spatial intensity.

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## 9. Public Input and Community Reports

- **What it does:** Collects reports from road users about dangerous or confusing road segments.
- **Use:** Supplements analytical methods.
- **Prioritization:** Based on frequency and seriousness of complaints.

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## 10. Composite Index or Scoring System

- **What it does:** Combines multiple factors (crash frequency, rate, severity, volume) into one composite score.
- **Example Scoring:**
  - Crash Frequency: 40%
  - Crash Rate: 30%
  - Severity Index: 20%
  - Public Complaints: 10%
- **Prioritization:** Rank based on total weighted score.

## Crash Reduction Capabilities and Countermeasures:

Crash reduction capability refers to the **estimated percentage decrease in crashes** (or crash severity) expected after applying a specific **safety countermeasure**.

**Example:** Installing a roundabout at an intersection may reduce fatal and injury crashes by 60% compared to a stop-controlled intersection.

## **Crash Reduction Countermeasures/Road safety design elements,**

1. **Roundabouts**
  - Replace traditional intersections to eliminate angle and head-on collisions.
  - Reduce conflict points and force lower vehicle speeds, improving safety.
2. **Rumble Strips (Edge/Center Line)**
  - Alert inattentive or drowsy drivers when they leave their lane.
  - Reduce run-off-road and head-on crashes significantly.
3. **Improved Street Lighting**
  - Enhances nighttime visibility for drivers and pedestrians.
  - Reduces crashes in poorly lit areas, especially at intersections.
4. **Pedestrian Crosswalk Enhancements**
  - Includes markings, raised tables, and flashing beacons at crossings.
  - Improves pedestrian visibility and driver compliance.
5. **Speed Humps and Calming Measures**
  - Physically reduce vehicle speeds in residential or high-pedestrian areas.
  - Lower speed leads to fewer and less severe crashes.
6. **Dedicated Turn Lanes(Islands)**
  - Separate turning traffic from through lanes at intersections.
  - Reduce rear-end and angle-type collisions by improving traffic flow.
7. **Median Barriers (e.g., cable or concrete)**
  - Prevent vehicles from crossing into oncoming traffic lanes.
  - Reduce fatal head-on collisions on divided highways.
8. **Traffic Signal Upgrades**
  - Includes signal timing, phasing, and visibility improvements.
  - Reduces intersection-related crashes and improves traffic efficiency.
9. **High Friction Surface Treatment (HFST)**
  - Applies specialized pavement to high-risk curves or intersections.
  - Improves traction, especially in wet conditions, reducing skidding crashes.
10. **Guardrails and Crash Cushions**
  - Protect vehicles from steep drops, obstacles, or fixed objects.
  - Absorb impact energy and reduce crash severity.
11. **Better Signage and Pavement Markings**
  - Improve driver awareness with clearer visual guidance.
  - Especially effective in curves, work zones, and merging areas.
12. **Road Safety Audits (RSA)**
  - In-depth safety assessments by expert teams on existing or planned roads.
  - Helps detect risks proactively and recommend targeted solutions.
13. **Access Management**
  - Controls location and number of driveways/intersections on a corridor.
  - Reduces conflict points and crash frequency on arterial roads.
14. **Pedestrian Overpasses and Underpasses**
  - Completely separate pedestrian movement from vehicle traffic.
  - Eliminate pedestrian-vehicle conflict and enhance safety.
15. **Speed Cameras and Automated Enforcement**
  - Detect and penalize speeding or red-light violations automatically.
  - Discourages risky behavior and reduces speed-related crashes.

## Methods to Estimate Crash Reduction Capabilities of Countermeasures:

### 1. Before-and-After Studies

- **Description:** Compares the number of crashes at a location before and after implementing a countermeasure.
- **Approach:**

$$\text{Crash Reduction}(\%) = \frac{C_{\text{before}} - C_{\text{after}}}{C_{\text{before}}} \times 100$$

### 2. Empirical Bayes (EB) Method

- **Description:** Adjusts before-and-after studies by factoring in expected crash frequency using reference (similar) sites.
- **Approach:** Combines actual crash data with predicted crashes from statistical models.

### 3. Crash Modification Factors (CMF)

- **Description:** Uses known CMFs from prior research to estimate crash changes.
- **Formula:**

◦  $\text{Expected Crashes After} = \text{Current Crashes} \times \text{CMF}$  Example:  
CMF = 0.6 → 40% expected crash reduction

### 4. Predictive Crash Modeling (Statistical Regression)

- **Description:** Uses models (e.g., Poisson or Negative Binomial regression) to predict crashes based on road features, volume, and control type.

### 5. Cross-sectional Studies

- **Description:** Compares crash rates across multiple sites, some with and some without a specific countermeasure.

### 6. Meta-Analysis / Literature Review

- **Description:** Uses published research studies to derive average crash reduction values.

### 7. Expert Judgment and Engineering Evaluation

- **Description:** Based on experience, crash diagrams, and field investigations.

## Common Road Safety Countermeasures and Their Crash Reduction Capabilities

| Countermeasure                            | Crash Type Targeted           | Typical Crash Reduction (%) |
|-------------------------------------------|-------------------------------|-----------------------------|
| Roundabout Installation                   | Angle, Head-on                | 60–80%                      |
| Rumble Strips (Center/Edge Line)          | Run-off-road, Head-on         | 20–40%                      |
| Improved Street Lighting                  | Night-time crashes            | 20–40%                      |
| Installing Guardrails                     | Run-off-road                  | 25–45%                      |
| Speed Humps / Calming Measures            | Speed-related, Pedestrian     | 40–70%                      |
| Signalizing Intersections                 | Right-angle, Rear-end         | 10–30%                      |
| Converting Signals to Roundabouts         | Angle, Fatal crashes          | 60–90%                      |
| Adding Turn Lanes at Intersections        | Left-turn/Rear-end            | 15–55%                      |
| Improved Signage and Markings             | All crash types               | 10–20%                      |
| Pedestrian Crosswalk Enhancements         | Pedestrian crashes            | 20–50%                      |
| Median Barriers (e.g., cable barriers)    | Cross-median, Head-on crashes | 50–90%                      |
| High Friction Surface Treatment (HFST)    | Wet surface crashes, Skidding | 30–70%                      |
| Access Management (e.g., driveway limits) | Rear-end, Angle crashes       | 15–30%                      |

- **Note:** Actual reduction percentages depend on site conditions, traffic volume, and crash history.

## **Effectiveness/Role Of Safety Design Features:**

### **1. Roadway Geometry Improvements**

Improving road geometry such as alignment, lane width, and shoulder space helps enhance driver control and visibility. These adjustments reduce sharp curves and blind spots, minimizing crash potential. Smoother transitions between road sections improve traffic flow. Roads designed with proper curvature and grade reduce speeding and skidding. Overall, this can reduce crashes by 30–40%.

### **2. Median Barriers**

Median barriers are installed on divided highways to prevent vehicles from crossing into oncoming traffic. They are especially effective in reducing head-on collisions, which are often fatal. Types include concrete, steel, or cable barriers. These barriers are low-maintenance and highly durable. Studies show they can reduce cross-median crashes by up to 90%.

### **3. Rumble Strips**

Rumble strips are grooved patterns on the road that create noise and vibration when driven over. They alert distracted or drowsy drivers who unintentionally drift from their lane. Typically placed on the shoulder or centerline of highways, they prevent run-off-road and head-on crashes. These strips are cost-effective and require minimal maintenance. Crash reductions of 20–80% have been observed.

### **4. Roadside Safety Treatments**

Improving the roadside environment involves removing or shielding fixed objects like trees, poles, or steep slopes. Guardrails and crash cushions absorb impact energy and prevent rollovers. Providing a clear recovery zone allows vehicles to regain control safely. These measures reduce the severity of run-off-road crashes. Crash reductions range between 30% and 45%.

### **5. Traffic Calming Measures**

Traffic calming involves physical design strategies to reduce vehicle speed in sensitive zones. Common features include speed bumps, raised pedestrian crossings, and road narrowings. These are vital in residential, school, and urban areas to enhance pedestrian safety. Slower speeds result in fewer and less severe crashes. Effectiveness shows crash reduction rates of up to 60%.

### **6. Pedestrian and Bicycle Facilities**

Dedicated paths, sidewalks, and crossings ensure the safety of non-motorized road users. These features separate pedestrians and cyclists from vehicular traffic, reducing conflict points. Signalized crossings, overpasses, and warning signage further enhance safety. Proper lighting and markings also play a role. Such facilities can reduce crashes involving pedestrians/cyclists by up to 50%.

## **7. Roundabouts**

Roundabouts are circular intersections that replace traditional traffic signals or stop signs. They reduce the number and severity of conflict points, especially right-angle and head-on collisions. Traffic moves continuously, which lowers speed and delays. Roundabouts are especially effective in reducing fatal and injury crashes. Crash reductions up to 90% have been reported for severe cases.

## **8. Access Control**

Controlling access to highways and arterial roads minimizes interruptions and conflict points. This includes limiting driveways, merging lanes, and using frontage roads. Reducing uncontrolled access points lowers the risk of rear-end and side-impact crashes. Access control also improves traffic flow efficiency. Studies show a 25–40% reduction in crash frequency.

## **9. Signage and Road Markings**

Proper signage and clear lane markings are crucial for communicating road rules to drivers. Signs warn about curves, speed limits, pedestrian zones, and upcoming intersections. Road markings guide vehicle movement and improve night time visibility. Well-maintained signs and paint reduce driver confusion and reaction time. These can lead to crash reductions of 10–30%.

## **10. Street Lighting**

Good lighting enhances visibility for drivers, especially at intersections, curves, and pedestrian crossings. It helps detect obstacles, road users, and signage during night time or adverse weather. Lighting reduces accidents caused by low visibility or surprise hazards. It is especially effective in urban areas with high pedestrian traffic. Night-time crash reduction rates range from 20% to 50%.

## **Accident Reconstruction:**

Accident reconstruction is a scientific process used to analyze motor vehicle collisions, determining the sequence of events, contributing factors, and potential causes. It involves a systematic approach, combining data collection, analysis, and sometimes computer modeling, to recreate the crash and understand what happened.

### **Objectives of Accident Reconstruction:**

- **Determining the cause of the accident:** Identifying factors such as driver error, vehicle malfunction, or environmental conditions.
- **Establishing the sequence of events leading up to the collision:** Reconstructing the actions of the vehicles and drivers involved.
- **Determining the speed and impact point of the vehicles:** Understanding the dynamics of the collision.
- **Assessing the role of different contributing factors:** Evaluating the impact of visibility, road conditions, and other elements.

### **Accident Reconstruction Procedure:**

#### **1. Preliminary Investigation and Data Collection:**

- **Scene Documentation:**  
This involves photographing and videotaping the accident scene, including vehicle positions, debris, and any relevant markings like skid marks.
- **Vehicle Examination:**  
A thorough inspection of the vehicles involved is crucial, noting damage, mechanical condition, and any pre-existing issues.
- **Evidence Gathering:**  
Collecting physical evidence such as tire marks, fluid spills, and any damaged components is essential.
- **Document Review:**  
Examining police reports, witness statements, medical records, and other relevant documentation provides valuable context.
- **On-board Diagnostics:**  
Data from the vehicles' Event Data Recorders (EDR), also known as "black boxes," can be downloaded to analyze vehicle speed, braking, and other parameters.
- **Interviews:**  
Speaking with witnesses, drivers, and first responders helps to gather firsthand accounts of the event.

#### **2. Data Analysis and Reconstruction:**

- **Applying Scientific Principles:**

Accident reconstruction experts use principles of physics, engineering, and mathematics to analyze the collected data.

- **Creating Scale Diagrams:**

Detailed diagrams of the accident scene, vehicle positions, and relevant measurements are created.

- **Computer Modeling and Simulation:**

Sophisticated software can be used to simulate the crash, allowing experts to test different scenarios and refine their understanding of the events.

- **Testing Theories:**

Various potential scenarios are tested using the collected data and analysis to determine the most likely sequence of events.

- **Determining Speed and Impact:**

Accident reconstruction can determine vehicle speeds at impact, angles of collision, and the point of initial contact.

### **3. Presentation and Reporting:**

- **Formal Report:**

A comprehensive report summarizing the findings, including the methodology, analysis, and conclusions, is prepared.

- **Visual Aids:**

Diagrams, animations, or 3D models of the accident can be created to help illustrate the findings and make them more understandable.

- **Expert Testimony:**

Reconstruction experts may be called upon to provide expert testimony in legal proceedings, explaining their findings and analysis to a judge or jury.

### **4. Recommendations & Preventive Measures**

Based on the reconstruction, investigators may recommend improvements—for example, adjustments to road design, vehicle safety features, or driver policies—to prevent future accidents.

### Accident Studies and records:

The various steps involved in traffic accident studies are collection of accident data, preparation of reports, location file and diagrams.

(1) Collection of accident data:

The collection of accident data is the first step in the accident ~~data~~ study.

The details to be collected are:

(1) General: Date, time, persons involved in the accident and their particulars, classification of accident like fatal, serious, minor etc.

(2) Location: Description and details of the location of accident.

(3) Details of vehicles involved: Registration number, make and description of the vehicles, loading details, vehicular defects.

(4) Nature of accident: Condition of vehicles involved; details of collision and pedestrians or objects involved, damages, injuries casualty etc.

(5) Road and traffic conditions: Details of road geometrics, whether the road is straight or curved, surface characteristics such as dry, wet or slippery etc. Traffic condition - type of traffic, traffic density etc.

(6) Primary causes of accident: Various possible causes and the primary cause of the accident.

(7) Accident costs: The total cost of the accident computed in terms of rupees of the various involvements like property damages, personal injuries and casualties.

## (2) Accident report:

- The accident should be reported to police authorities who would take legal actions especially in more serious accidents involving injuries, casualties or severe damage to property.
- Accident report of the individuals involved may be separately taken.
- The accident data should be collected as given above and the accident report is prepared with all facts which might be useful in subsequent analysis, claims for compensation etc.

## (3) Accident records:

The accident records are maintained giving all particulars of the accidents, location other details. The records may be maintained by means of location files, spot maps, collision diagrams and condition diagrams.

(1) Location files: These are useful to keep a check on the location of accident and to identify points of high accident incidence. Location files should be maintained by each police station for the respective jurisdiction.

(2) Spot maps: Accident location spot maps show accidents by spots, pins or symbols on the map. A map of suitable scale. Say,  $1\text{cm} = 40\text{ to }60\text{m}$ , may be used for spotting urban accidents.

### (3) Condition diagram:

- A condition diagram is a drawing to scale showing all important physical conditions of an accident location to be studied.
- The important features generally to be shown in this diagram with suitable dimensions marked there in an roadway limits, curves, kerb lines, bridges, culverts trees and all details of roadway conditions, obstruction to vision, property lines, signs, signals etc.
- There are standard symbols used in showing various details.
- The Condition and collision diagrams may be combined together in a single sketch, if necessary.

### (4) Collision diagram:

- These are diagrams showing the approximate path of vehicles and pedestrians involved in the accidents.
- Collision diagrams are most useful to compare the accident pattern before and after the remedial measures have been taken.

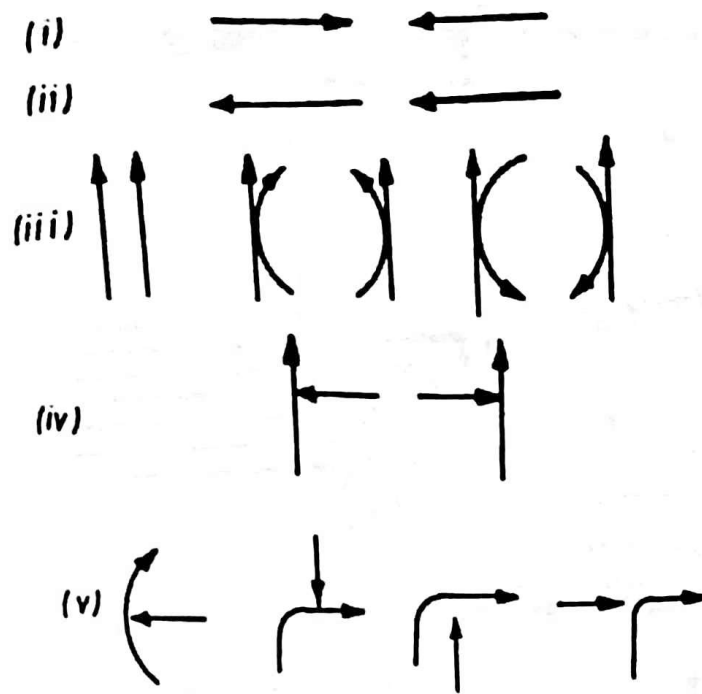


Fig. 18.1

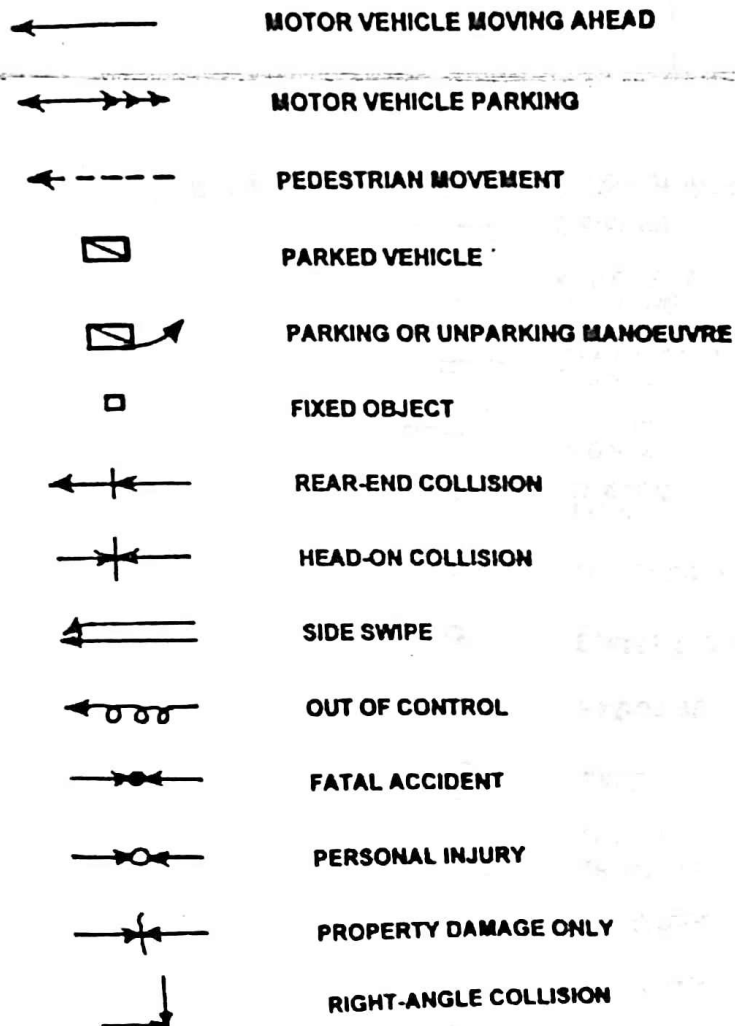


Fig. 18.2. Symbols employed in collision diagrams.

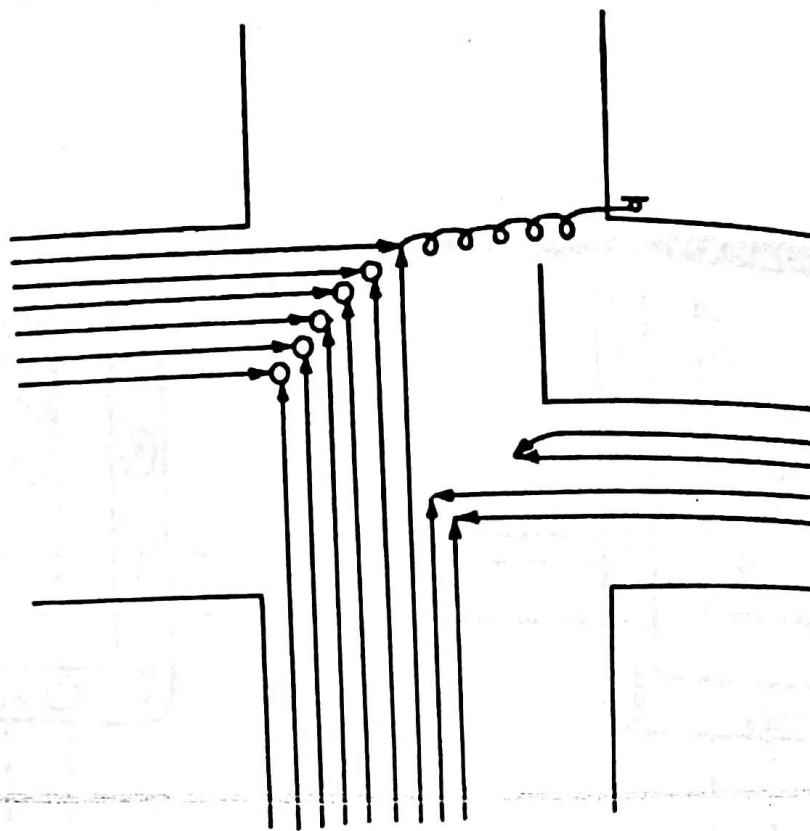


Fig. 18.3. Typical collision diagram at a junction.

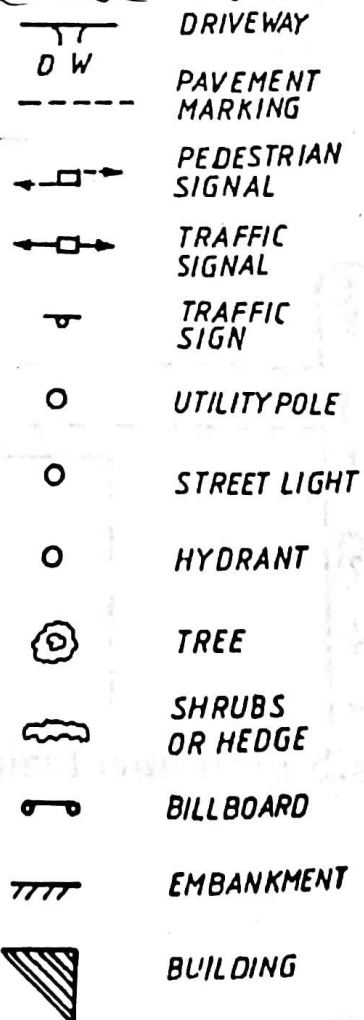


Fig. 18.4. Symbols for condition diagrams.

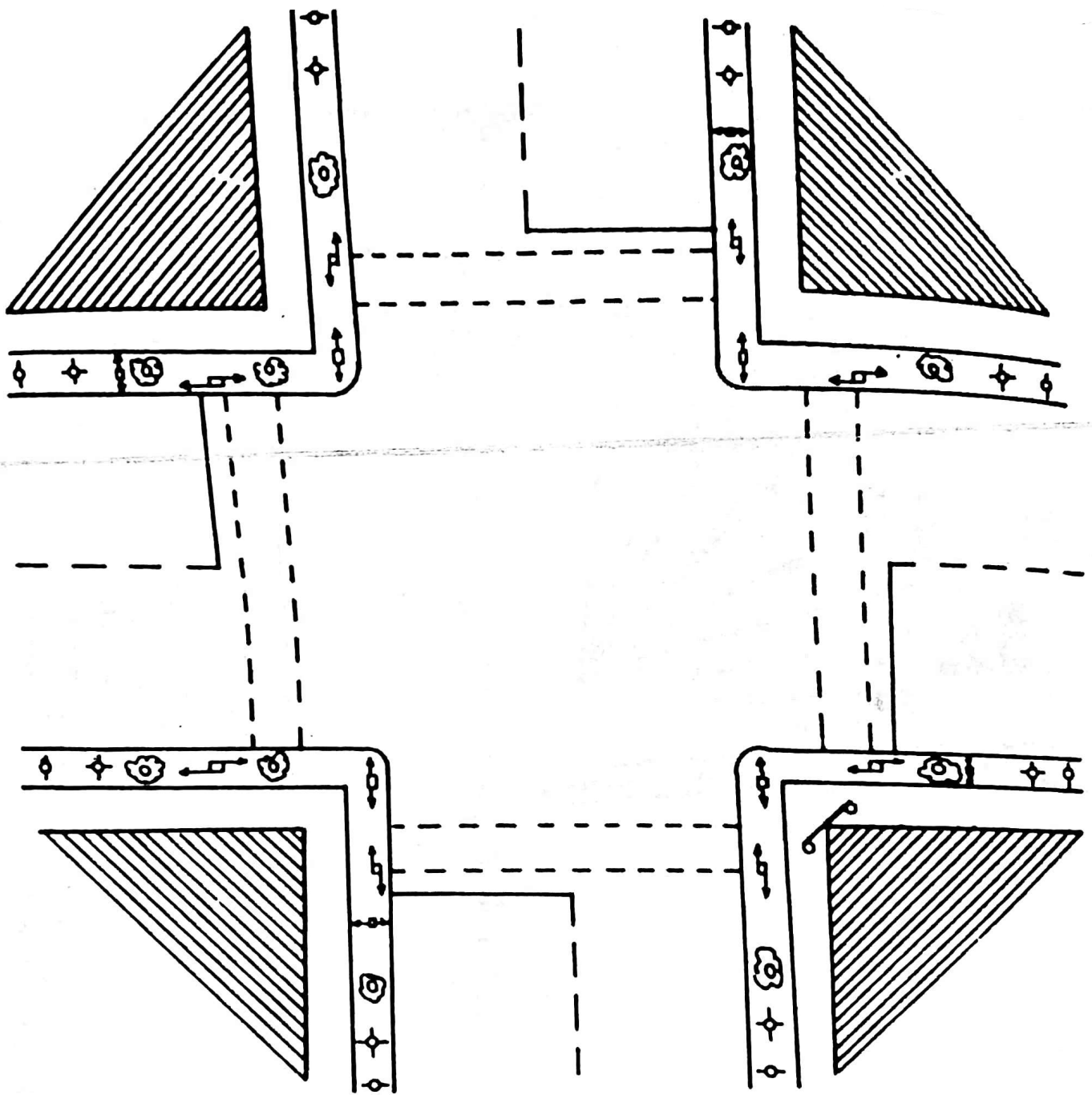


Fig. 18-5. Typical condition diagram.

Accident investigations & studies therefore may be carried out scientifically in the following three stages:

- (1) Accident investigations.
- (2) Analysis of Individual Accidents.
- (3) Statistical Analysis of Accidents.

(1) Accident Investigations:

The following investigations may be carried out to enable analysis of the accident on a scientific basis:

(a) Recording General observations:

- Measurement of length of skid marks due to partial and full skidding.
- Recording the relative positions of vehicles and objects involved in the accident and collision diagram supplemented with photographs.
- Details of accident, injuries and damages.
- Condition of pavement surface, shoulders and other surface through which the vehicles involved in the accident have moved, environment condition
- Conditions diagram of accident locations with relevant measurements and dimensions.

(b) Driver Tests:

- Analysis of breath of drivers involved in the accident for alcohol content. In case the driver is dead, tests may be conducted on the spinal fluid for estimating the alcohol content, if any.
  - Tests on driver characteristics such as reaction time, distance judgement, angle of clear vision etc.
- (43) If the accident has occurred during night, glare vision and glare recovery tests are to be conducted on the driver in addition to the above tests

### 3) Skid Resistance of the Pavement Surface:

The avg skid resistance of the road surface along the skid marks should be measured under the prevailing condition of the surface soon after the accident, using portable pendulum type skid resistance tester or any other suitable equipment.

### 4) Vehicle Tests:

- Tests on the condition of brakes and steering of the vehicles involved.
- Tests on essential accessories and general condition of the vehicles involved in the accident.
- Characteristics and details of the vehicles and other objects involved and the cross section details of the collapsed members.

### (5) Probable cause of the accident:

Assessment of the probable causes of the accident, its type, site conditions, position of the vehicles and other objects involved and other existing conditions.

### (6) Cost Analysis:

- Estimation of the cost of accident by working out the cost involved for the following items:
- Injuries and fatalities of persons involved.
  - Damages to the vehicles.
  - Property damages.
  - Other consequences including traffic delay.
  - Investigations and legal proceeding.

## (2) Statistical analysis of accidents:

- The statistical analysis of road accidents help to assess the effectiveness of various measures to decrease the accident rate; thus the analysis estimates the relative safety of road stretches or zones.
- As the mobility increases the probability of accidents also increases.
- The number of accidents are found to increase with the number of road users or the number of vehicles and pedestrians on the roads.
- The effectiveness of traffic engineering, enforcement and educational measures may therefore be judged from the changes in the annual accident rates, rather the total number.
- The accident rate may either be expressed in terms of the number of vehicles and population or the vehicle movements and the population.
- Relative accident rate may also be expressed in terms of various factors of the roadway and traffic.
- By statistical study of accident occurrence at a particular road or location or a zone of study for a long period of time, it is possible to predict with reasonable accuracy the probability of accident occurrence per day or the relative safety of different classes of road users in that location.
- The reliability of the statistical analysis of accident depends mainly on the reliability of the accident data, obtained from the accident records.

### Analysis of individual traffic accidents:

Some of the typical modes of vehicular accidents are:

- 1) A moving vehicle collides with a parked vehicle.
- 2) Two vehicles approaching from different directions collide at an intersection.
- 3) Head-on-collision of two vehicles approaching from opposite directions.
- 1) A moving vehicle collides with a stationary object like an electric pole, tree or a rigid structure.

### **Question Bank for Module 1**

1. Illustrate various objectives of accident studies.
2. Enumerate various data to be collected during accident study.
3. Illustrate various methods/Sources available to collect accident data.
4. Explain the steps involved in Assessment of Road safety.
5. Describe the key performance indicators used in road safety assessment. How do these indicators influence road safety planning and policy development?
6. List and explain the major contributing factors/Causes to road traffic crashes/accidents.
7. Enumerate various Methods/Techniques to Identify and Prioritize Hazardous Locations and Elements.
8. Explain various crash reduction counter measures/Road safety elements/Design features.
9. Enumerate various Methods to Estimate Crash Reduction Capabilities of Countermeasures.
10. Analyze the Role/Effectiveness of road safety design features in accident prevention.
11. Explain the objectives and Procedure of accident Reconstruction.
12. Explain with a neat sketch Condition and Collision diagram used in accident studies.
13. Write a note on accident report.